Script-based simulation

Xor.hdl

```
CHIP Xor {
    IN a, b;
    OUT out;

    PARTS:
    Not (in=a, out=nota);
    Not (in=b, out=notb);
    And (a=a, b=notb, out=aAndNotb);
    And (a=nota, b=b, out=notaAndb);
    Or  (a=aAndNotb, b=notaAndb, out=out);
}
```

Xor.tst

```
load Xor.hdl;
set a 0, set b 0, eval;
set a 0, set b 1, eval;
set a 1, set b 0, eval;
set a 1, set b 1, eval;
```

test script = sequence of commands to the simulator

Benefits:

- Automatic testing
- Replicable testing.

Script-based simulation, with an output file

Xor.hdl

```
CHIP Xor {
    IN a, b;
    OUT out;

    PARTS:
    Not (in=a, out=nota);
    Not (in=b, out=notb);
    And (a=a, b=notb, out=aAndNotb);
    And (a=nota, b=b, out=notaAndb);
    Or  (a=aAndNotb, b=notaAndb, out=out);
}
```

Xor.tst

```
load Xor.hdl,
output-file Xor.out,
output-list a b out;
set a 0, set b 0, eval, output;
set a 0, set b 1, eval, output;
set a 1, set b 0, eval, output;
set a 1, set b 1, eval, output;
```

test script

Xor.out

```
| a | b |out|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |
```

Output File, created by the test script, as a side-effect of the simulation process

The logic of a typical test script

- Initialize:

  Loads an HDL file

  Creates an empty output file

  Lists the names of the pins whose values will be written to the output file

- Repeat:

  `Set` (inputs) , `eval` (chip logic) , `output` (print)